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## Mini Project Report on

## “AUTOMATED PATIENT EDUCATION SYSTEM”

Submitted in partial fulfillment of the requirements for the  
**First Semester of the Bachelor of Engineering Degree**, towards the completion of the **Mini Project** under the **Innovation & Design Thinking Laboratory**,  
Department of Basic Sciences.

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## DEPARTMENT OF BASIC SCIENCES



## CERTIFICATE

This is to certify that the File Structures mini project entitled “Automated Patient Education System” has been successfully carried out by Srividya M.S(L-58), Vaibhav Ranjan(L-59) and Syed Aausaf(L-60) by bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Innovation & Design Thinking Laboratory, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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## Abstract

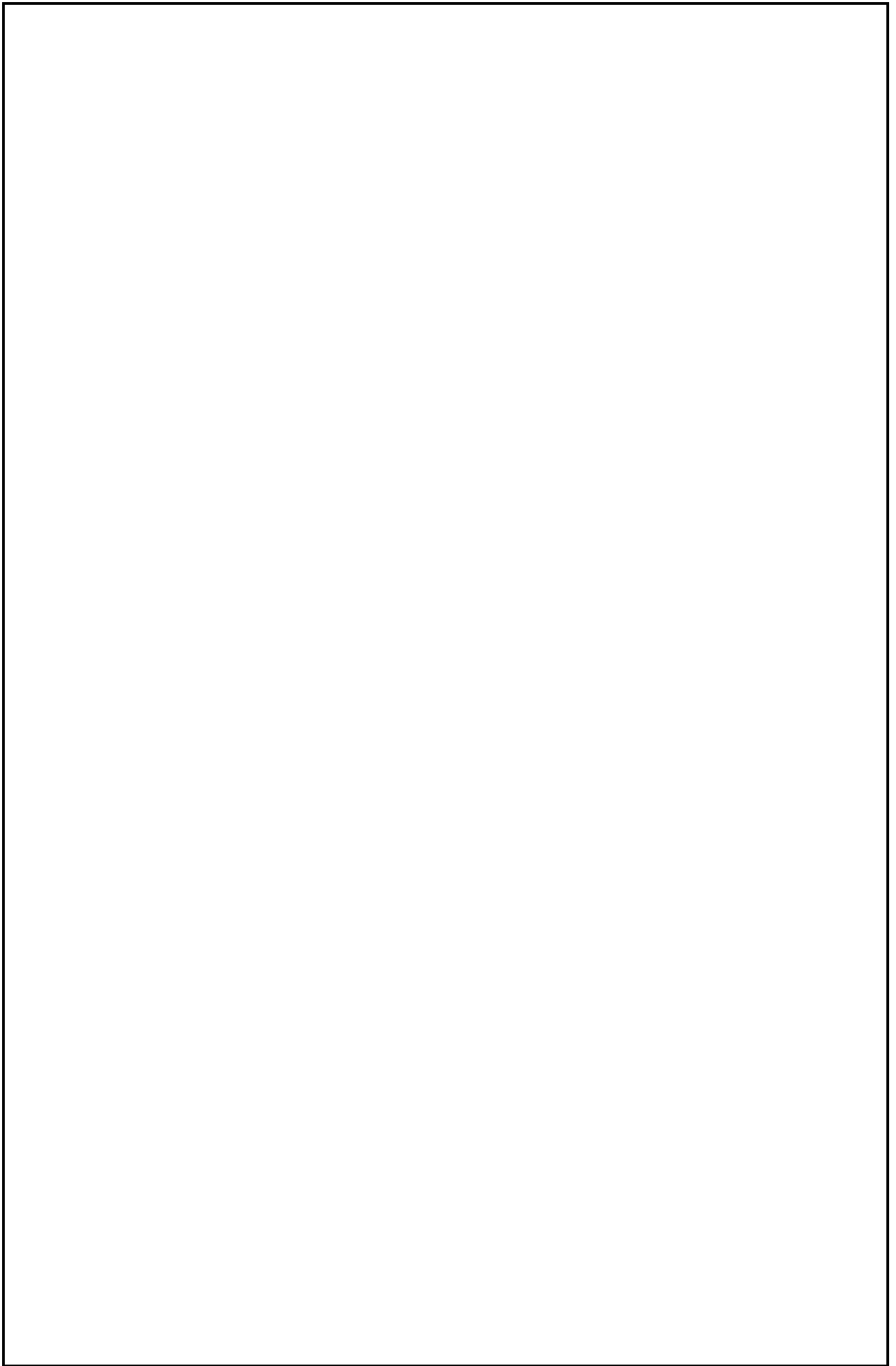
The Automated Patient Education System (APES) is a digital healthcare platform designed to provide accurate, consistent, and easy-to-understand medical information to patients. The system delivers education about diseases, symptoms, diagnosis, treatment options, preventive measures, and lifestyle management through an automated interface. By using multimedia content such as text, images, voice narration, and interactive quizzes, APES improves patient understanding and engagement. This system reduces the workload of healthcare professionals while ensuring patients receive reliable medical guidance anytime and anywhere.

## Keywords

- ✓ Automated Patient Education System,
- ✓ Digital Healthcare,
- ✓ Artificial Intelligence,
- ✓ Patient Awareness,
- ✓ Disease Education,
- ✓ Preventive Care,
- ✓ Health Informatics,
- ✓ Smart Healthcare

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## **Chapter 1**

### **INTRODUCTION**

Patient education plays a vital role in effective healthcare delivery. Proper understanding of diseases and treatments helps patients follow medical advice accurately, resulting in better health outcomes. However, traditional patient education methods such as verbal explanations and printed pamphlets are often insufficient due to limited consultation time, language barriers, and lack of follow-up. The Automated Patient Education System addresses these challenges by providing a digital platform that educates patients in a structured and user-friendly manner. The system ensures consistency in information delivery and allows patients to revisit content whenever needed. By automating education, healthcare institutions can improve service quality and patient satisfaction.

#### **1.1 Brief history of attendance system**

Early patient education relied entirely on face-to-face interactions and printed materials. With the advent of computers, hospitals introduced digital records and basic informational websites. Recently, advancements in AI, cloud computing, and mobile applications have enabled automated, intelligent, and multilingual patient education platforms.



## **1.2 Modern Biometric system**

Modern systems integrate:

- ✓ Web and mobile applications.
- ✓ Multimedia content (text, images, videos, voice).
- ✓ AI chat-based personalization.
- ✓ Real-time data access.
- ✓ Interactive assessments.

## Chapter 2

### Problem Statement

#### 2.1 Description:

Patients often lack proper understanding of their medical conditions due to:

- Limited consultation time with doctors.
- Language barriers.
- Complex medical terminology.
- Absence of follow-up educational support.

This results in medication errors, poor lifestyle management, and repeated hospital visits.

#### 2.2 Challenge Statement:

To design and implement a **user-friendly, automated, and scalable patient education system** that delivers accurate medical information in an understandable format while supporting different diseases and patient needs.

- ❖ Accuracy of Medical Information
- ❖ Language Barriers
- ❖ Digital Literacy of Users
- ❖ Content Maintenance
- ❖ Integration with Hospital Systems
- ❖ Voice Assistance Limitations
- ❖ User Engagement
- ❖ Legal and Ethical Concerns

## Chapter 3

### 3.1 Design Thinking Process

#### 1. Empathize:

Interviews were conducted with patients, doctors, and nurses. Patients expressed confusion about treatment plans and preventive care after consultations.

#### 2. Define:

Key requirements identified:

- Simple language explanations.
- Disease-specific education.
- Voice and visual support.
- Accessibility anytime, anywhere .

#### 3. Ideate:

Multiple ideas were generated including mobile apps, kiosks, chatbots, and web platforms. The selected solution was a **web based automated patient education system**.

#### 4. Prototype:

A functional prototype is web based development with diseases, modules, multimedia ,content, quizzes, and voice narration was developed.

#### 5. Test:

Testing showed improved patient understanding, engagement, and satisfaction

## **3.2 Methodology**

### **1.Requirement Analysis:**

Identify patient education needs, common diseases, and chatbot functionalities.

### **2.System Design :**

Design the website interface, chatbot interaction flow, and system architecture.

### **3.Website Development:**

Develop web pages to display disease information such as symptoms, diagnosis, treatment, precautions, and diet.

### **4.Chatbot Integration:**

Implement a chatbot to answer patient queries using a predefined medical knowledge base.

### **5.Backend & Database Development:**

Store disease content and chatbot responses in a database and manage them using backend logic.

### **6.Voice Assistance Integration:**

Provide optional text-to-speech support for better accessibility.

### **7.Testing:**

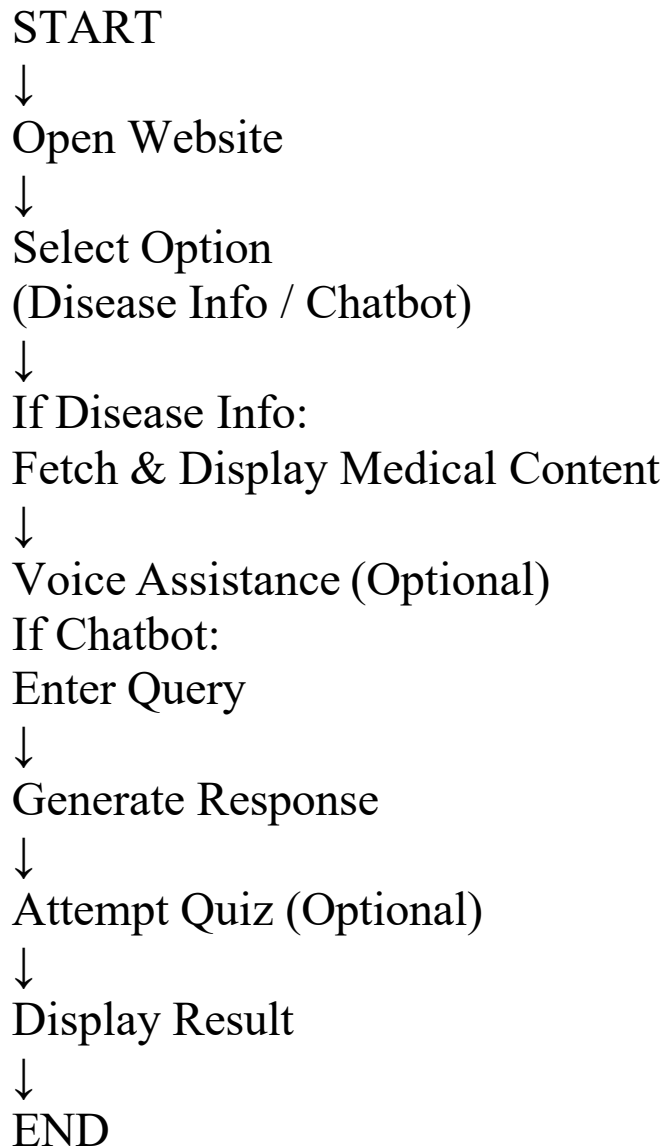
Test website navigation, chatbot responses, and overall system functionality.

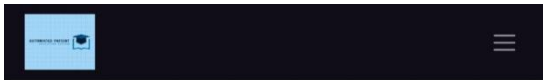
### **8.Deployment & Maintenance:**

Deploy the system and update medical content and chatbot knowledge regularly.

## PROTOTYPE DESCRIPTION:

### FLOW CHART:



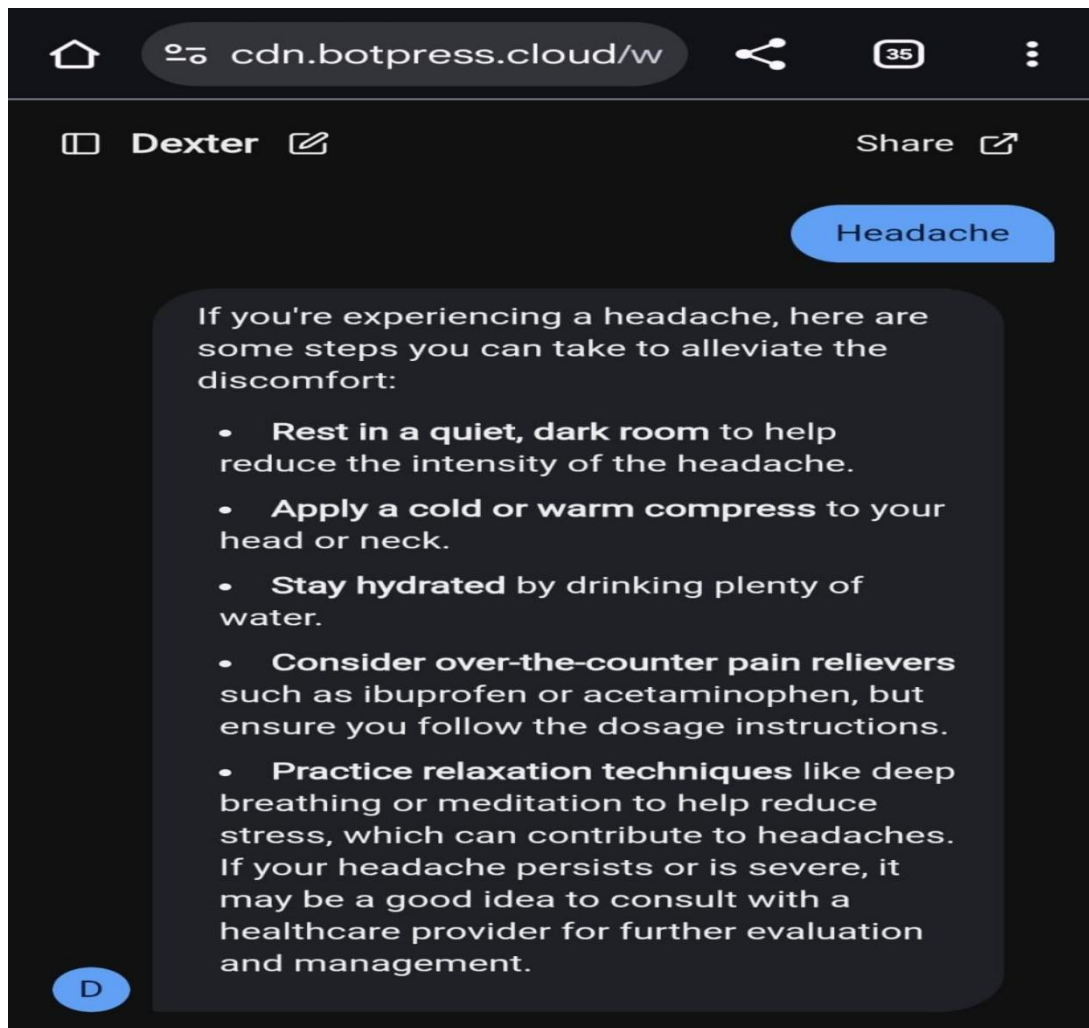


**DEXTER** 

Dexter is a medical AI chatbot designed to be

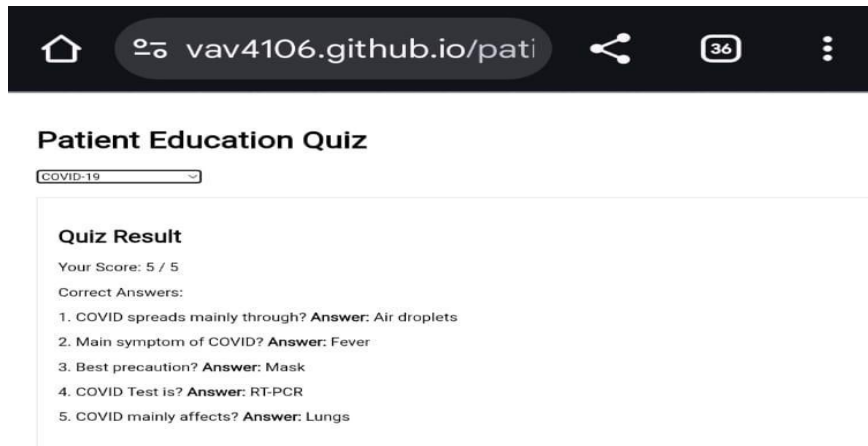
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**Screenshot 1:** The prototype showcases an AI chatbot named Dexter that provides automated patient education through a simple mobile interface. It delivers clear information on diseases, symptoms, prevention, treatment, and lifestyle guidance using conversational interaction. The system improves patient awareness, accessibility, and understanding of healthcare information without requiring direct hospital visits.



### **Screenshot 2:**

This screen demonstrates the AI chatbot Dexter providing instant patient guidance for a reported symptom (headache). The chatbot delivers clear, step-by-step self-care advice, basic medication guidance, and safety recommendations, supporting quick patient education and encouraging medical consultation if symptoms persist.



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## Prototype Description – Patient Education Quiz

For Example:

This prototype screen displays an interactive patient education quiz for COVID-19. It shows the user's score and correct answers, helping assess understanding of symptoms, transmission, prevention, and diagnosis. The quiz reinforces learning, improves awareness, and validates patient knowledge through instant feedback.



## **Chapter 4**

### **Results and Analysis**

#### **User Testing & Feedback**

##### **Sample:**

Participants: 10 patients and 2 healthcare staff

##### **Quantitative Results**

1. Understanding level increased by ~70%
- 2.Reduced consultation clarification time
- 3.Quiz accuracy improved after education

##### **Qualitative Feedback**

Patients: “Easy to understand and helpful.”

Doctors: “Reduces repetitive explanation work.”

Nurses: “Improves patient confidence.”

## Chapter 6

### **Conclusion & Future Work:**

#### **Future Work :**

- AI-based personalized recommendations.
- Multilingual support .
- Mobile application integration .
- Chatbot-based interaction.
- Integration with hospital management systems.

#### **Conclusion :**

The Automated Patient Education System is an effective solution that delivers personalized and consistent health information to patients efficiently. By using technology, it helps patients better understand their medical conditions, treatments, and preventive care, leading to improved compliance and health outcomes. This system also eases the workload of healthcare professionals by automating routine education tasks. Despite challenges like ensuring data privacy and making the system accessible to all, its benefits in empowering patients and improving the quality of care make it a valuable addition to modern healthcare.

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5. Computers in Biology and Medicine

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